LMU Munich
Frauenlobstraße 7a
D–80337 München
Department for Informatics

Bachelor Thesis

Spatial Masking of Vibrotactile Signals on the Hand

Kira Telegina

Ki.Telegina@campus.lmu.de

Course of Study:	Medieninformatik
Examiner:	Prof. Steinbach, Prof. Butz
Supervisor:	Lars Nockenberg, M.Sc. Research Assistant
Commenced:	October 28, 2024
Completed:	March 17, 2025

“Spatial Masking of Vibrotactile Signals on the Hand”
2025-03-17

© 2025 Kira Telegina

This work is licensed under a Creative Commons Attribution-ShareAlike 4.0 License (CC BY-SA 4.0):
http://creativecommons.org/licenses/by-sa/4.0/

Typesetting: PDF-LaTeX 2$_\varepsilon$
Character Count: 44372

Summary

This thesis examines the phenomenon of vibrotactile masking, focusing on how a masking signal impacts the perception of another signal across various frequencies. This effect is essential for improving haptic feedback systems in fields like virtual reality, assistive technology, and human-computer interaction.

For better understanding of this phenomenon, a psychophysical experiment was conducted using the three-alternative forced-choice (3AFC) method and an adaptive staircase process. Participants completed multiple trials in which they were tasked to detect the presence of a target vibration (maskee) under different masking conditions. The experiment contained different combinations of the masker and maskee frequencies to determine how the interaction influenced the participants' perception.

The results confirmed that frequency influences the effect of masking but does not always follow a linear relationship. Some results demonstrated strong masking at frequencies which are close to each other, but others showed the deviations.

These findings highlight the complexity of vibrotactile perception and the importance of better experimental approaches in future research. On a larger scale, this study contributes to the development of more precise haptic feedback systems by illustrating the impact of masking effects on tactile perception.

Abstract

This study examines vibrotactile masking thresholds at different positions on the hand using a psychophysical experiment. This is achieved through a masker-maskee combination to analyze the boundaries of human vibrotactile perception. The experiment utilizes a three-alternative forced-choice technique (3AFC) and will analyze various stimulus frequencies and intensities. The goal is to determine the masking threshold and to identify the point at which the maskee becomes undetectable due to the masker, using combinations of white noise and pure tones as masker-maskee pairs. The amplitude of the maskee will be iteratively adjusted based on the participant's responses. The experimental procedure also includes actuator calibration and determination of the perceptual threshold as a reference level. A key aspect of the study is the experimental design, which aims to effectively find the relevant information about the spectral and spatial dependencies of masking. Additionally, the results should be analyzed in detail to identify factors that influence masking. These findings contribute to a better understanding of haptic perception, which will help to optimize haptic codecs.

Contents

1 Introduction **11**
- 1.1 Relevance of the Research . . . 11
- 1.2 Aim of the study . . . 11
- 1.3 Research Tasks . . . 12

2 Related Work **13**
- 2.1 Masking in audiology and tactile research . . . 13
- 2.2 Analysis of Previous Studies . . . 13
- 2.3 Contribution to the Experiment . . . 14

3 Study Design **15**
- 3.1 Implementation of the Software . . . 15
- 3.2 Procedure . . . 20
- 3.3 Adjustment of the audio signal . . . 20
- 3.4 General principles . . . 21
- 3.5 Factors Influencing Vibrotactile Perception . . . 21

4 Results **23**
- 4.1 Raw data . . . 23
- 4.2 Evaluation . . . 24

5 Discussion **27**
- 5.1 Threshold Analysis . . . 27
- 5.2 Influencing Factors . . . 27

6 Conclusion **29**

Bibliography **31**

List of Figures

1 Python software. GUI. 16
2 Descending staircase corresponding to a masker-maskee combination. 16
3 Windowing of white noise. Blue line is the orignial signal, and orange line is a windowed signal. 18

4 Results of masking experiments. It shows the masking thresholds for 5 frequencies with masker and one without masker for 3 subjects. 23

1 Introduction

1.1 Relevance of the Research

Modern haptic feedback technologies play an important role in developing interfaces for virtual reality, medicine, robotics, and human-machine interaction systems. These technologies facilitate the transmission of information through tactile sensations, which is especially important in situations where visual or auditory feedback is restricted. However, successfully implementing these technologies requires a deep understanding of the limitations and characteristics of human vibrotactile perception.

One key challenge in this field is the study of masking, a phenomenon where one vibrational signal (the masker) reduces or completely hides the perception of another signal (the maskee) [15]. This effect is essential for the design of haptic feedback devices that should transmit the information precisely and reduce the conflicts between signals that could affect the correct transmission. This research focuses on perception thresholds in scenarios where one vibrational signal interferes with another one, aiming to find necessary parameters that will optimize tactile information delivery.

Previous studies analysed the effects of frequency and intensity on masking, but the impact of how spatial positioning influences perception thresholds has not been fully assessed. Addressing these gaps will contribute to a more deep understanding of vibrotactile masking and its potential applications in haptic technology development.

1.2 Aim of the study

This study aims to explore perception thresholds for vibrational signals under tactile masking conditions. The research achieves this goal by utilizing a specially prepared software that identifies the boundaries of human vibrotactile perception through a reproduction of masker-maskee combinations. The results will contribute to the improvements of haptic technologies, facilitating the development of better and highly accurate interfaces. Furthermore, this study also analyzes human tactile perception, making it valuable for both practical applications and theoretical research.

A psychophysical experiment examines the spatial and spectral aspects of vibrotactile masking on the hand. The experiment measures masking thresholds at two fingers, analyzing how the masker influences the perception of the maskee across different frequencies and intensities. The goal is to determine the point where the maskee is no longer felt because of the masker, and also to provide a detailed analysis of the factors affecting vibrotactile perception, especially in terms of spatial and spectral interactions. Additionally, the findings contribute to the optimization of haptic codecs by providing insights into the limitations of human vibrotactile perception.

A better understanding of how vibrotactile masking works has benefits across many fields. First, the study provides new ideas about human haptic perception, especially regarding the ability to detect and differentiate tactile signals under unusual conditions. This knowledge plays an important role in designing more effective haptic interfaces and devices, including those used in virtual reality, assistive technologies, and wearable systems. By identifying how spatial positioning and frequency influence the masking, this research improves the accuracy of tactile feedback systems. Additionally, optimizing haptic codecs, which encode and decode tactile signals, enhances user experiences in technologies relying on advanced tactile feedback, making

them more responsive and immersive [9]. The findings of this study have direct implications for fields like neuroscience and engineering, providing future innovations in human-computer interaction and assistive technologies.

1.3 Research Tasks

To achieve the goals of this study, the research considers several important aspects. First, it reviews the theoretical principles of vibrotactile perception and masking. Second, the custom software that was made for this experiment generates white noise and sinusoidal signals and implements the three-alternative forced-choice (3AFC) method for participant testing. This psychophysical technique is widely used for improving measurement accuracy and reducing response biases. Introduced in sensory and perception research, it asks participants to identify the correct stimulus among three options, reducing the chances of random guessing compared to two-alternative forced-choice (2AFC) tasks [10]. This approach is especially useful for threshold estimation in auditory, visual, and tactile perception studies. After preparing the software and equipment, participants interact with the system in an experiment in order to evaluate their ability to perceive vibrational signals both with and without masking. The collected data are then analyzed to determine perception thresholds and examine the factors influencing vibrotactile masking. Based on these findings, the study concludes and proposes recommendations to improve haptic feedback technologies.

This study explores the theoretical background of vibrotactile perception and masking effects and describes technical aspects of the experiment's, including software development, functional requirements, study design, participant testing process, and methods of data collection. Finally, the analysis of results focuses on the factors that influence masking and compares the findings with previous studies.

2 Related Work

2.1 Masking in audiology and tactile research

Masking is a phenomenon in which one signal reduces or completely suppresses the perception of another. In audiology this explains why it is difficult to distinguish quiet sounds in a noisy environment [11]. Vibrotactile masking follows a similar principle - when two vibrational signals are applied to the skin simultaneously, one may reduce or completely block the perception of the other [15]. So appears a masking effect. Research shows that this effect depends on several factors. Frequency plays an important role: vibrations at higher frequencies tend to mask other high-frequency signals more effectively, while low-frequency signals primarily mask those within the same range [2]. Another crucial factor is intensity, as a stronger masker makes it harder to perceive the maskee.

This phenomenon is widely examined in the field of haptic feedback, particularly for applications in virtual reality, medicine, and assistive communication systems. A deeper understanding of vibrotactile masking allows to optimize algorithms responsible for transmitting tactile information, improving the accuracy and effectiveness of haptic feedback devices.

2.2 Analysis of Previous Studies

An important area of research is the study of how certain vibrations interfere with the perception of others, a phenomenon similar to effects that appear in auditory perception.

A study by Rahul Chaudhariu et al. (2015) provides empirical evidence for a broadband masking effect in vibrotactile perception [5]. The results indicate that vibrations with frequencies close to a strong masker experience greater suppression than those at more distant frequencies. The study also found that masking strength depends on both signal intensity and frequency spectrum, which helps identify the frequencies at which signals are most noticeable. These findings are essential for optimizing haptic feedback devices, such as medicine systems and virtual reality applications, where masking effects need to be considered when processing and compressing tactile data. Furthermore, this research introduced a vibrotactile texture codec designed to improve data compression without significantly affecting perceived quality. By leveraging the masking effect, this codec reduces the amount of transmitted tactile data while maintaining essential sensory information. Such approaches are particularly useful in applications where transmission and storage of tactile data are necessary.

Another important study by Verrillo et al. (1983) examined the perception thresholds of vibratory signals under masking conditions [15]. The research explored how the spatial location of the masker (on one or two areas of the hand) and the frequency characteristics of both the masker and maskee influence perception. The study found that vibrotactile perception is based on two distinct receptor systems: Pacinian corpuscles, which are sensitive to high frequencies (300 Hz), and non-Pacinian receptors, which respond to low frequencies (13 Hz). These systems operate independently, meaning that masking primarily occurs within the same system. For example, a high-frequency signal is more effectively masked by another high-frequency signal.

2.3 Contribution to the Experiment

The reviewed studies provide essential theoretical background for the current experiment. They have valuable information about vibrotactile masking, variables that are important for experiment and factors influencing perception thresholds.

The study by Chaudhari et al. (2015) demonstrates experimental setup by describing how frequency relationships between masker and maskee affect perception. As a result, different frequency combinations are tested to analyze their impact on the strength of masking. Additionally, the research on haptic data compression emphasizes the importance of measuring perception thresholds accurately, because this data can be used to optimize future haptic encoding techniques.

The work of Verrillo et al. (1983) provides physiological context, confirming that vibrotactile masking occurs primarily within the same receptor system. This finding influences the choice of frequencies used in the experiment to ensure strong masking effects where they are most likely to occur. Moreover, the study shows the importance of spatial factors in masking, informing about the placement of vibrotactile stimuli across different hand positions.

These studies provide key features for the experiment, including the choice of frequencies, testing of different intensities and importance of spatial positioning. Current research aims to refine the understanding of vibrotactile perception thresholds and contribute to the development of more effective haptic feedback technologies.

3 Study Design

3.1 Implementation of the Software

This section describes the functionality, adaptive algorithm, and user interface of the software developed for the experiment. The program is responsible for generating vibrational stimuli, optimization of experimental settings to measure vibrotactile perception thresholds, managing trial conditions, and automatically recording participant responses. The program aims to provide signal accuracy, algorithm adaptability, and interaction for participants.

The program generates two types of signals: band-limited white noise, which serves as the masking signal (masker), and a sinusoidal signal with adjustable frequency and amplitude, which acts as the masked signal (maskee). If the maskee overlaps with the masker, it can lead to an interference, distorting the perception. Band-limited white noise which spreads across a wider frequency range provides a more stable masking effect by reducing noticeable changes in transmission of vibrations [3]. This approach allows for more precise manipulation of masking effects while preventing direct frequency interactions between the masker and maskee.

To measure perception thresholds, the program applies the three-alternative forced-choice (3AFC) method with a 1-up 3-down (1U-3D) adaptive staircase procedure. Each trial presents three consecutive stimulus intervals. The maskee appears in only one randomly selected interval per trial, while the masker (when present) remains in all three intervals. Randomized choice of interval ensures unbiased results and prevents participants from guessing the answer and improves the reliability of the collected data.

The experiment consists of two parts. In the first part, participants are presented with one condition, defined by a maskee frequency 200 Hz with an initial amplitude of -20 dB. Throughout this phase, the maskee is played without a masker. The second part introduces a masker with changed maskee frequencies. In this phase, the masker frequency stays unchanged and makes 200 Hz. The frequency of the maskee ranges from 140 to 260 Hz. In both phases, the maskee appears in only one randomly selected interval out of the three, ensuring that participants cannot predict its position. In the second phase, the masker remains present in all three stimulus intervals, but the maskee still appears in only one of them. Randomizing the interval placement eliminates predictability and ensures unbiased data. Additionally, the sequence of conditions follows a randomized order to prevent learning effects.

During this experiment participants interacted with a graphical user interface (GUI) (Fig. 1). The interface displays instructions, stimulus intervals, and selection buttons. After perceiving the maskee, participants select the interval where they detected it, and the program saves the response. The system stores all responses in a CSV file according to a trial number, selected interval, current dB level, and maskee amplitude.

The stimulus duration is setted to 1 second, with 500 ms pauses between intervals. A 1-second break occurs between trials to allow participants to focus. These settings follow recommendations from previous studies, such as those by Chaudhari et al [5].

The adaptive staircase method gradually adjusts the maskee's amplitude based on participant responses. Three consecutive correct answers reduce the amplitude, making the signal harder to detect, while a single incorrect one increases amplitude, making it easier to detect. This adjustment guides participants toward their perception threshold, where detection occurs with a 50% probability [7]. In each part of the experiment the combination without masker was evaluated by the subjects one time and with masker five times, tracing

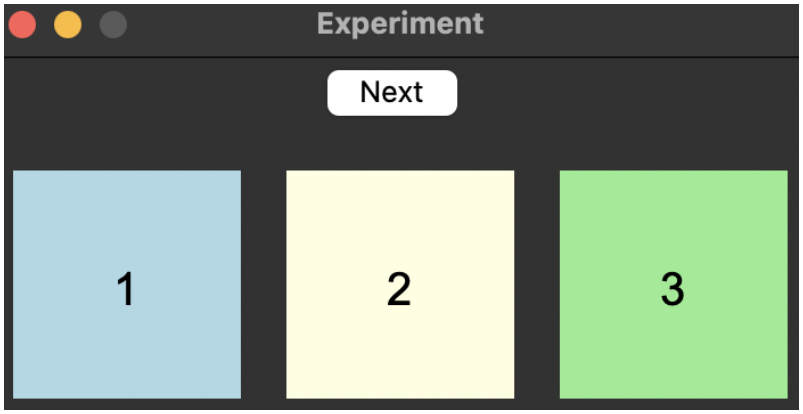


Figure 1: Python software. GUI.

both a descending staircase. For the descending staircase, the initial maskee amplitude was chosen to be well above the expected detection threshold level, so that the maskee was easily detectable at the beginning. The final threshold is calculated by averaging the values from all six staircase reversals. Additionally, the system allows for the manual configuration of initial parameters, including frequency, amplitude and duration, to adapt different experimental conditions.

According to the paper from Chaudhari et al., the step size for adjusting the maskee's intensity was set to 8 dB to accelerate convergence toward the perception threshold [5]. With each reversal of the staircase, the step size was reduced by half until it reached a minimum of 0.5 dB (see Fig. 2). This final step size was chosen as an optimal balance between measurement precision and experimental efficiency. Once the step size reached 0.5 dB, it stayed fixed for the rest of the staircase. The staircase procedure terminated after three consecutive reversals (change of intensity from increasing to decreasing) at this minimum step size.

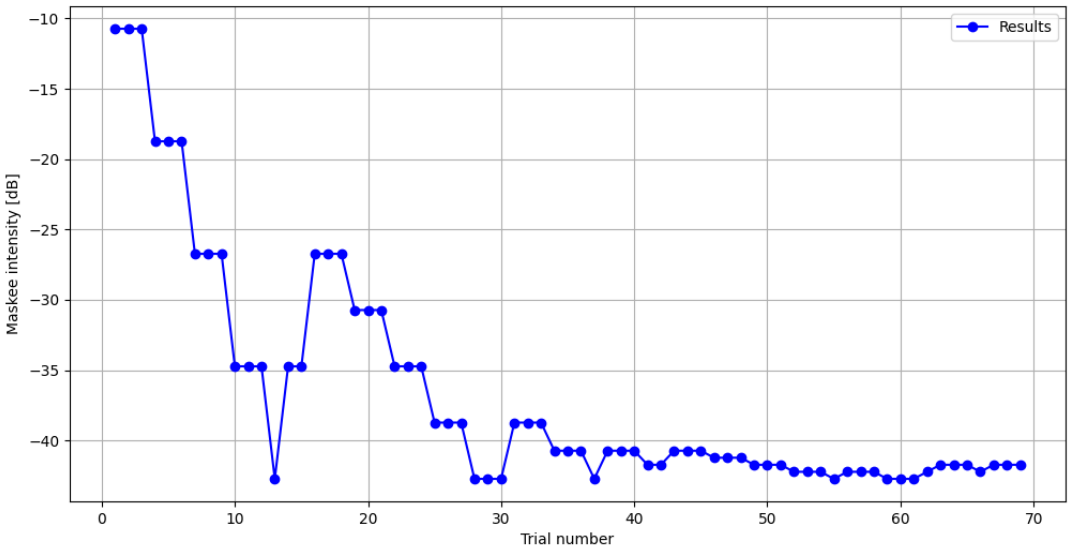


Figure 2: Descending staircase corresponding to a masker-maskee combination.

Before transitioning to trials with the masker, the program prompts participants to take a break. To optimize threshold measurement, the software retrieves previous participant data, filters relevant results and adjusts the starting amplitude for the next phase.

During the next phase, when trials with the masker begin, the script iterates through all parameter combinations. For each combination where the masker is played, the program extracts the participant's recorded threshold for the corresponding maskee frequency from the CSV-file. Based on this threshold, the initial amplitude of the maskee is adjusted by adding 45 dB, while the initial amplitude of the masker is set to the threshold value plus 25 dB. This adaptive adjustment ensures that the masker and maskee levels are appropriately scaled relative to the participant's detected threshold. This adjustment ensures that the perceived intensity remains approximately the same for all participants, creating optimal conditions for evaluation of masking effect. By scaling the masker and maskee levels relative to threshold, the experiment considers different vibrotactile perception, allowing for more consistent and comparable results across participants.

Finally, the program automatically collects and saves the experimental data. This includes the number of the trial, the participant's answer, the intensity value in dB, which is measured relative to 1 m/s^2, the current dB value after decreasing or increasing the amplitude, and the current amplitude of the maskee. The intensity value in dB is expressed relative to 1 m/s^2 and means that all amplitude values are normalized against this standard, ensuring consistency across measurements and allowing for meaningful comparisons between different trials and participants [17]. Additionally, a file is created to store the data of the parameters. The data is saved in a format convenient for further analysis, including CSV-files and PNG images for graphs. A separate folder with a unique name is created for each experiment, containing all the data of the specific participant.

The ability to adjust parameters helps enhance the perception threshold estimation, while organized data storage allows for an objective analysis of masking effect. These features improve the reliability of the experiment and provide a deeper understanding of vibrotactile perception.

3.1.1 Sine Wave and White Noise

Accurate signal processing plays an important role in ensuring the reliability of experimental results in vibrotactile perception studies. The experiment utilizes two distinct types of signals: a sinusoidal signal (maskee) and filtered white noise (masker). The sinusoidal signal represents the target stimulus that participants must detect, while white noise serves as a masking stimulus that hinders perception.

The generation of the sinusoidal signal begins with determining its amplitude. The system converts the decibel level into a linear amplitude and scales it relative to a reference level to maintain consistency across trials [12]. A time sequence is created using the sampling rate and stimulus duration, ensuring the precise generation of the signal at the required maskee frequency.

To ensure that all sine waves maintain a consistent intensity, the system applies normalization using root mean square (RMS) scaling [13]. This step adjusts the signal's amplitude to a predefined RMS value, preventing fluctuations in perceived intensity caused by natural variations in signal amplitude.

Precise signal processing plays an important role in ensuring the reliability of experimental results in vibrotactile perception studies. The white noise serves as a masking stimulus, which interferes with the perception of the maskee signal. To maintain experimental conditions, the masker is filtered, normalized and amplitude is scaled. These steps ensure that the masker effectively interacts with the maskee signal and minimizes unpredictable changes.

The unfiltered white noise extends across a broad spectrum, so a bandpass filter is applied to hide the relevant frequency range [4]. This filter removes unwanted frequency components, allowing only those within the defined lower and upper cutoff limits to pass. Restricting the masker's frequency content ensures that it influences the perception of the maskee without introducing additional interference outside the targeted frequency range. Once the noise passes through a filtering process, the system applies normalization to adjust its intensity to a predefined root mean square value [13]. This step ensures that all masker signals maintain

a consistent amplitude across different trials, preventing deviations in perceived intensity. The finalized signals were filtered and normalized to ensure that all stimuli were presented under controlled and repeatable conditions.

3.1.2 Windowing

Windowing plays an important role and ensures that vibrational stimuli are presented smoothly and without distortions. By gradually increasing and decreasing the signal's amplitude at the beginning and end, it prevents sudden changes that could introduce unwanted noise or interfere with how participants perceive the stimuli [16]. This technique ensures that all signals are processed in the same way.

At the initial stage of software development, windowing was applied to test and ensure the quality of the generated signals. This technique smoothed the onset and offset of vibrations, preventing sudden changes that could introduce distortions or unwanted noise. By gradually increasing and decreasing the amplitude at the signal's boundaries, windowing minimized unwanted irregularities such as spectral leakage and distortions [16]. To verify the accuracy of signal processing, the system generated a comparison between the original and windowed versions of both the sine wave maskee and the white noise masker (Fig. 3). This comparison ensured that windowing effectively smoothed the transmission of the signal while keeping the original quality of it.

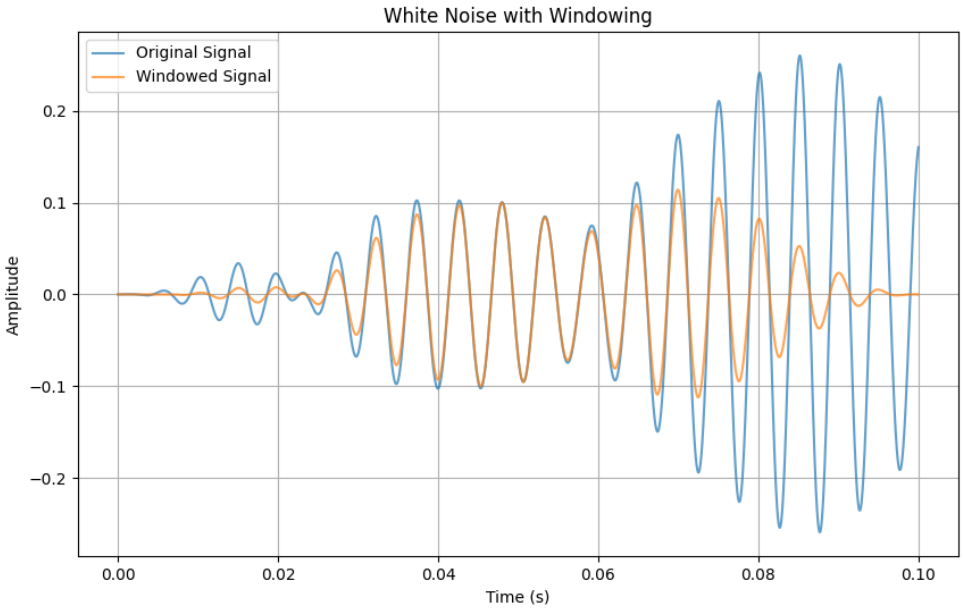


Figure 3: Windowing of white noise. Blue line is the orignial signal, and orange line is a windowed signal.

In the experiment, windowing was important for improving the accuracy of perception measurements. Without it, abrupt signal transitions could create unwanted vibrations in the actuators, potentially influencing participant responses. By applying windowing, the system ensured that all stimuli remained stable and controlled, allowing for a more precise determination of perception thresholds. To ensure accurate signal presentation, a calibration process was conducted using an accelerometer, which measured the acceleration amplitude of the actuator [14]. The calibration procedure corrected the actuator's frequency response and

adjusted the signal scaling, so that 0 dB in the software corresponded to an amplitude of 1 m/s^2 [17]. This step minimized mismatches between the digital signal and the actual vibrations perceived by participants, improving the reliability of the experiment.

3.1.3 Bandpass Filter

Bandpass filtering helps to keep the masker's effect on the maskee precise and controlled by limiting the masker to a specific frequency range. Since white noise naturally contains a wide range of frequencies, using it without filtering could interfere with perception in unpredictable ways [4]. The bandpass filter removes unnecessary frequencies, allowing the masker to interact with the maskee without frequencies outside the selected range. The system automatically adjusts the filter's frequency range based on the masker's central frequency. It sets lower and upper limits to prevent interference from frequencies that are either too low or too high. The filter's width controls how much of the frequency spectrum is retained.

Applying a bandpass filter makes perception measurements more precise by ensuring that only a specified band of frequencies contributes to masking. By defining the filter parameters in the system, each trial supports consistent conditions, reducing variability between participants and making the results more precise.

3.1.4 Normalization

Normalization ensures that all stimuli in the experiment have the same intensity, preventing the influence of differences in signal strength on perception or results. This adjustment allows participants to focus on detecting the stimuli without being affected by unexpected changes in amplitude.

A key part of normalization is Root Mean Square (RMS), which measures a signal's overall energy rather than just its peak amplitude [13]. The system calculates the RMS value by averaging the squared amplitude values and then taking the square root. This method ensures that even if two signals have different peak values, their perceived intensity remains the same.

Using RMS normalization improves signal quality by preventing distortions such as clipping, which occurs when a signal is too strong, or loss of details, which happens when a signal is too weak to be detected. In the experiment, normalization helps to keep all trials stable and accurate, ensuring that any changes in perception come from the masking effect and not from differences in signal strength.

3.1.5 Threshold Determination

A threshold is the lowest intensity at which a participant can securely detect a stimulus. It is a key measure of how sensitive a person's sensory system is, whether in hearing or touch perception [8]. Finding the threshold helps researchers understand perceptual limits and how they vary between individuals. Setting of stimulus levels based on these thresholds ensures that signals are neither too weak to detect nor too strong, because this can influence responses. Threshold measurement is especially important in masking studies, where one stimulus makes another harder to perceive. By comparing detection thresholds with and without a masker, the experiment determines how much the masker affects perception.

By combining bandpass filtering, normalization, and threshold determination, the experiment keeps signal transmission precise and controlled, allowing for an accurate analysis of vibrotactile perception.

3.2 Procedure

1) Subjects: Three subjects participated in the psychophysical experiments – 2 female and 1 male. Their ages ranged from 23 to 30, with an average of 26 years. All of them were right handed and had participated in haptic psychophysical experiments before. None of them reported having any discomfort that would affect their sensory-motor ability.

2) Experimental Setup: In the experiment, participants interacted with a tactile system that included a Python program, an actuator and two sensors. The program generated three intervals and transmitted vibrations through the actuator to the two sensors, where participants placed their fingers. The subjects wore acoustic noise-canceling headphones to mask out the external noise for better concentration. They interacted with the experiment's GUI through the mouse. Their task was to determine in which of the three intervals the maskee was present and select the corresponding option on the screen.

3) Method: The program is designed to generate, play, and process signals using an adaptive algorithm. These factors are essential for ensuring accurate perception threshold measurements and maintaining consistency across all trials.

The first stage of development focuses on implementing of the white noise and sinusoidal signal generation. White noise covers all frequencies equally, creating a static background, while the sinusoidal signal adjusts its frequency and amplitude. Both signals have equal duration and are combined into consecutive sequences, where one of the three intervals randomly contains the maskee. The three-alternative forced-choice (3AFC) ensures that participants cannot foresee where the maskee will appear. The duration of each interval and the pauses between them remains fixed, keeping the experimental conditions consistent. The program also has an adaptive algorithm for adjusting experimental parameters during the experiment. The amplitude of the sinusoidal signal changes based on participant responses - decreasing after three correct detections and increasing after incorrect ones.

A randomization in trials, adaptive parameter adjustments, and automated data collection ensures the objective and reproducible results. The adaptive staircase method helps to identify individual perception limits and threshold, while automated data processing reduces the potential error in recording results. These features improve the reliability of the experiment and provide valuable insights into the understanding of vibrotactile perception mechanisms.

3.3 Adjustment of the audio signal

The white noise was normalized using a bandpass filter. It limits the noise to a particular frequency range, reduces frequencies outside of it and prevents it from masking frequencies that are not relevant to the experiment [4]. This ensures that the masker only affects specific parts of the frequency spectrum, making the results more controlled.

The amplitude of both the masker and maskee was measured using the RMS formula. RMS provides an average energy level of the signal, making it more representative of how loud the sound actually perceives to human ears [13]. A range was chosen from -1 to 1. If the volume exceeded this range, the program displayed a warning about high signal level. Initially, the RMS value for the masker and maskee was 4.8457705980514066e+89, which exceeded the allowed range. Later, the value of the amplitude changed to a supposed range, which means that its perceived loudness was the same despite possible differences in the signal reproduction.

In the beginning for all intervals the initial masker amplitude is -20 dB and initial maskee amplitude has the value of -10 dB. This is a moderate volume level that provides a sufficient difference in perception between the masked and masking signals. If the signal level is too low, it becomes harder to distinguish against the

background of white noise, while if it is too high, distortions may occur due to system overload [1]. Then during the experiment the initial amplitude is adjusted for each sequence: for maskee by adding 45 dB and for masker plus 25 dB.

3.4 General principles

Vibrotactile perception is the ability of the human sense of touch to detect and interpret mechanical vibrations transmitted through the skin. This process relies on specialized mechanoreceptors, primarily Pacinian corpuscles, which respond to high-frequency vibrations [15].

To accurately analyze perception thresholds, the experiment uses specific parameters of stimulation, including frequency, amplitude and its placement. These parameters were selected based on preliminary testing to provide optimal conditions for assessing sensitivity to different signal combinations. Frequency determines which receptors are activated, with Pacinian corpuscles responding most effectively to vibrations around 250–300 Hz [15]. Amplitude affects the perception threshold, as stronger vibrations are easier to detect, while weaker ones may be masked by other stimuli. The location and duration of vibration also influence perception, as different areas of the hand and body vary in receptor density.

Since human tactile perception is most sensitive to frequencies below 1000 Hz, extremely high masker frequencies are generally not effective for masking. Pacinian corpuscles, which are primarily responsible for detecting vibrations, have peak sensitivity around 250–300 Hz [5]. If the masker frequency is too high, it may not be efficiently processed by mechanoreceptors, reducing its interaction with the maskee and making it difficult to measure perception thresholds accurately.

A key measure in the analysis is the perception threshold, which defines the minimum intensity at which a participant detects a stimulus. The experiment determines the absolute threshold by using adaptive psychophysical methods, such as the staircase method. This approach continuously adjusts the maskee's amplitude based on participant responses, allowing the system to determine the exact intensity at which the stimulus becomes perceptible.

To evaluate masking effects, the system compares the threshold of the maskee with and without the masker. If the detection threshold increases when the masker is present, this indicates a strong masking effect. The method proves whether these differences are significant, proving that the results represent actual perceptual changes and not the random changes.

Beyond threshold comparisons, the analysis also examines how masking effects change across different frequencies and intensities. The data is visualized through threshold curves, which highlight the relationship between stimulus parameters and perception limits, producing precise and consistent results.

3.5 Factors Influencing Vibrotactile Perception

A key aspect of the experiment is the evaluation of how different factors influence vibrotactile perception thresholds. Participants tested different frequency parameters on different finger placement to analyse how these parameters affect masking effects and sensory detection.

The experiment requires participants to place their index and middle finger on tactile sensors to perceive vibrations. Since different areas of the skin have varying sensitivity levels, testing other fingers helps to determine whether vibrotactile masking effects will change based on finger placement. Different finger combinations, such as the thumb and index finger or the thumb and middle finger, may lead to varying masking effects. The interaction between two selected contact points could influence vibrotactile perception, as certain combinations might increase or weaken sensitivity to different frequencies [6]. The analysis of the

sensitivity of different fingers could provide insights into how tactile information is processed across multiple points of contact. However, this aspect was not explored within the scope of this study and opens an area for future research.

The experiment also examines how different parameters of frequencies influence the perception of the maskee under masking conditions. Since different receptors in the skin respond to specific frequency ranges [8], testing multiple frequencies allows to determine whether higher or lower frequencies are more sensitive to masking effects. For instance, high-frequency signals may be masked more effectively by high-frequency noise, while low-frequency signals may interact differently [2]. Understanding these relationships helps to improve models of vibrotactile perception and contributes to the development of more precise haptic technologies.

Not all frequency values provide meaningful experimental results. Preliminary testing was conducted to determine the most effective frequency ranges for measuring masking effects. Studies suggest that mechanoreceptors, particularly Pacinian corpuscles, are most sensitive to vibrations below 1000 Hz, making higher frequencies ineffective for producing strong masking effects [15]. As a result, a masker at this frequency and higher does not significantly interfere with the detection of the maskee, so it is not necessary for analysing masking effects.

To ensure consistency, the system defines the maskee and masker frequencies at the beginning of each experiment. This prevents variability between trials and ensures that comparisons remain valid. Fixed frequency parameters can accurately estimate how finger placements and frequency combinations affect perception thresholds avoiding unforeseen changes in stimulus properties.

Considering all these points, the experiment provides a more complex understanding of vibrotactile perception. Analyzing how masking effects differ across fingers and frequency ranges contributes to more effective haptic feedback development in fields such as virtual reality, assistive technologies, and human-computer interaction. These findings help optimize the transmission of tactile information, ensuring that haptic signals are delivered in ways that align with the natural sensitivity of the human hand.

4 Results

4.1 Raw data

On this graph you can see the

This study examines the relationship between threshold and maskee frequency based on the results of three participants, highlighting individual features in vibrotactile masking effects. The dataset consists of four experiments conducted with three participants, each completed several trials with different parameter settings but with the same maskee and masker frequency combinations. As depicted in Fig. 4, the data includes the measured thresholds (dB), the frequencies of the maskee signals (Hz), and whether the masker was present during the trial. The first participant completed two experiments, with the second experiment focusing only on the maskee frequency of 200 Hz. The second participant completed a single experiment, and the third participant also participated in one experiment. The results confirm that the masking effect depends on frequency and is influenced by individual sensitivity and experimental conditions.

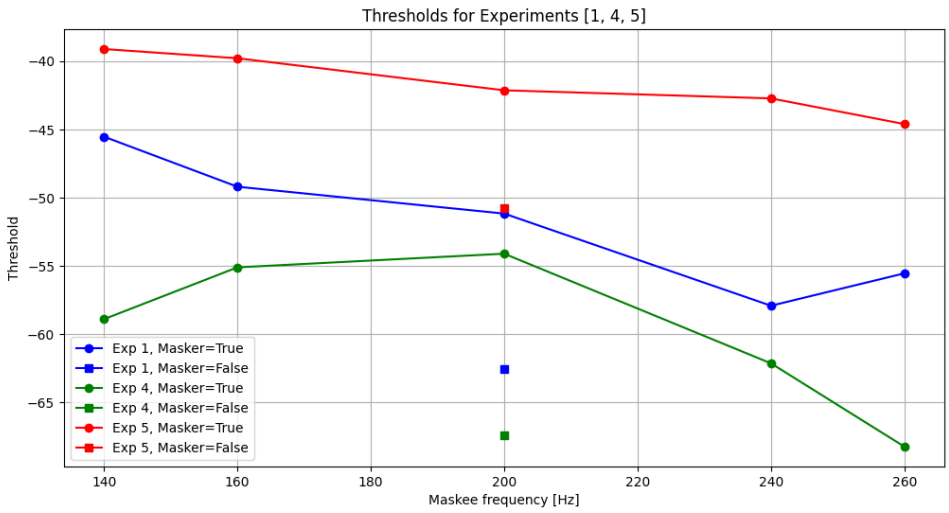


Figure 4: Results of masking experiments. It shows the masking thresholds for 5 frequencies with masker and one without masker for 3 subjects.

The first participant completed six trials with different maskee frequencies and masker settings. The perception threshold gradually decreased from 140 Hz to 240 Hz and the thresholds for the trials ranged between -45.52 dB and -62.53 dB. In the first experiment, the lowest threshold was observed at -62.53 dB when no masker was present. It means that the masker significantly increases the difficulty of detecting the maskee. With the masker at 200 Hz, the strongest masking effect appeared at 240 Hz and 260 Hz, where the threshold values reached -57.90 dB and -55.53 dB accordingly. It has increased from 240 Hz to 260 Hz, meaning either a reduction in masking effectiveness or an interaction between frequency sensitivity and masking intensity. These results confirm that the masking effect is not equal across frequencies. Instead of following a linear tendency, the threshold shows nonlinear behavior, reflecting the complex interplay between

the masker and maskee frequencies. Since this data comes from a single participant, individual differences may noticeably influence the results, emphasizing the need for further research with a larger sample size. The mean threshold value of this participant is -53.80 dB.

The same participant performed another experiment only for the maskee frequency of 200 Hz, aiming to obtain more precise measurements for this specific frequency. This experiment included both conditions: three trials with and one trial without a masker. By trials without a masker the threshold is -63.89 dB, which is very close to the value of -62.53 dB recorded in the first experiment under the same condition. It means that the participant's absolute threshold for detecting the 200 Hz maskee remained stable across trials, indicating minimal influence of learning effects or adaptation to the no-masker condition. When the masker at 200 Hz was introduced, the measured thresholds in the repeated experiment are -51.93 dB, -53.37 dB and -56.05 dB, with an average of -53.79 dB. In contrast, the first experiment recorded a threshold of -51.15 dB under the same masking condition. The wider range of values in the second experiment suggests that while the masking effect is generally present, it may vary across trials. These variations could be influenced by factors such as attentional changes and adaptation effects. The noticeable spread in threshold values indicates that three tests may not be enough to capture a precise and reliable masking threshold. Increasing the number of tests with the same frequency could help reduce diversity and provide a more consistent measurement of the masking effect at 200 Hz.

The second participant, who only completed one experiment, demonstrated an average threshold of -41.06 dB, with values ranging from -39.7 dB to -44.6 dB. The perception threshold gradually decreased from 140 Hz to 240 Hz. Compared to the first participant, this suggests differences in tactile perception and individual sensitivity to vibrotactile stimuli.

The third participant, who also completed one experiment, had the lowest overall thresholds -59.68 dB across all participants, showing higher sensitivity to vibrotactile stimuli and ability to detect the maskee at lower amplitudes. The results aligned more closely with those of the first participant, particularly in the stronger masking effects observed at 240 Hz and 260 Hz, where the threshold values are -62,12 dB and -68,23 dB. Compared to the second participant, the detection threshold without a masker was lower, measured at -67.42 dB for a maskee frequency of 200 Hz, which means that masking significantly raises the detection threshold. The threshold distribution for this participant indicates that masking effects were strongest at lower frequencies, in contrast to the broader variation in sensitivity seen in the second participant. Unlike the second participant, the third one showed a more consistent pattern across different maskee frequencies, suggesting either greater focus or a different sensitivity level to vibrotactile input.

The data highlights individual differences in vibrotactile perception. While the first and third participants showed a more predictable threshold pattern, the second participant's responses are more variable. This means that for the future analyses should be more participants to assess overall tendency and potential outliers.

4.2 Evaluation

The results demonstrate clear personal differences in vibrotactile sensitivity. The third participant was the most sensitive to the vibrations, followed by the first participant, and the second participant had the highest thresholds. The repeated experiment for 200 Hz maskee frequency helped the first participant to confirm that three tests are not enough to accurately determine the masking effect at this frequency. Additionally, comparing the results show that exposure duration or different variations of parameters may influence results. Adding more frequencies to check would be necessary to confirm these results and explain possible learning effects or adaptation.

A comparison of these results highlights several important discoveries. The second participant sequentially demonstrated higher thresholds, which means weaker masking effects compared to the first and third participants. This result points to individual differences in tactile sensitivity, which may be influenced by

factors such as mechanoreceptor distribution, finger placement, concentration, or attentional differences. While the 200 Hz masker had a clear masking effect, its influence on perception thresholds was not consistent across all participants. For two participants, the threshold at 200 Hz was positioned between their lowest and highest registered values, rather than at the peak, assuming that the masking effect depends on additional reasons and not only because of the frequency closeness.

The mean threshold values for each participant illustrate significant differences in their sensitivity to vibrotactile stimuli. For the first participant it is -53.80 dB, for the second is -43.07 dB, for the third is -60.31 dB. Different mean thresholds over all tests indicate that the second participant had higher thresholds, meaning that this participant needed a stronger stimulus to detect it, whereas the first and third participants showed lower thresholds, proposing larger sensitivity to stimulus.

The parameter without a masker was included in the experiment to establish the absolute threshold, which represents the lowest intensity at which a participant can detect a vibrotactile stimulus in the absence of interference. This serves as a measurement, allowing for a direct comparison between tests with and without a masker. By measuring the absolute threshold first, the experiment ensures that any increase in threshold when a masker is introduced is because of the masking effects rather than individual features in sensitivity. The difference between the absolute threshold and the threshold measured with a masker indicates how much the masking signal hinders perception.

The experiment explored vibrotactile masking thresholds under different frequency conditions by evaluating the limits of perception across three participants. The results confirm that masking increases perception thresholds, making vibrotactile signals more difficult to detect. Future research with a larger number of participants will help refine these findings and further explore the relationship between frequency, intensity, and spatial perception in vibrotactile masking.

5 Discussion

5.1 Threshold Analysis

Several factors may have contributed to variations in perception thresholds. Individual features in sensitivity to vibrotactile stimuli such as skin properties or finger placement played an important role. These variations could explain why one participant demonstrated higher thresholds than the others.

Masker intensity was another important factor. The influence of masker frequency on maskee perception varied among participants (See Fig. 4). The third participant's results confirmed the expectation that frequencies that are close to each other, for example maskee and masker frequency of 200 Hz, cause interference more strongly, as shown by the greater threshold shifts at other frequencies. The second participant demonstrated weaker masking effects, with relatively stable thresholds across frequencies, indicating that masker-maskee frequencies that are close to each other did not seriously increase perception thresholds. The first participant demonstrated intermediate results: masker effects appeared at certain frequencies, but in the repeated test with 200 Hz it was uncertain. These differences mean that frequency closeness plays a role in masking, but also depends on personal sensitivity and adaptation effects. However, deviations in threshold values across tests indicate that masking effect depends on frequency, for example, one participant with certain frequencies is more sensitive to suppression than others.

The results show that masking effects were strongly pronounced at 240 Hz and 260 Hz, corresponding to the peak sensitivity of Pacinian corpuscles (250 Hz) [5]. The results for the 200 Hz maskee frequency showed that close masker-maskee frequencies do not always produce a strong masking effect, especially when the masker has the same frequency as the maskee. These findings confirm that vibrotactile masking does not follow a strictly linear pattern but is influenced by combinations between masker and maskee frequencies.

Although this experiment primarily focused on observation of frequency and intensity, the placement of stimuli may have influenced the perception. Since two fingers have different sensitivity, their positioning could have influenced detection, because the distance also plays a role: the closer two points are, the harder it is to detect the stimulus.

5.2 Influencing Factors

The results show that masking thresholds depend on many factors like frequency and intensity, including concentration, attention, cognitive factors and the choice of experimental parameters. Each participant demonstrated unique perception thresholds, likely due to different susceptibility, skin properties and tactile adaptation. Cognitive and attentional factors also influenced results. Some participants may have been more engaged, leading to lower thresholds, while others had reduced sensitivity due to tiredness or due to getting used to the process itself.

Finger placement and contact pressure contributed to threshold differences, because they could have altered vibration transmission, affecting perception. Despite calibration, minor deviations in actuator positioning may have affected the signal transmission. Additionally, randomized trial sequences may have influenced detection ability, as the effect of the masker at the previous trial could have affected later responses.

These findings emphasize the complexity of vibrotactile perception and the need to check other frequencies to obtain more outcomes. Future studies should combine various conditions and consider the personal differences to improve the understanding of the masking effects that depend on frequency.

The analysis confirms that masking effect is frequency-dependent, with stronger masking observed at 240 Hz and 260 Hz. The 200 Hz maskee frequency together with a masker of the same frequency showed the high thresholds only by one participant. Additionally, the different features of tactile sensitivity contributed to variability of results of all participants, pointing that other factors besides frequency play a role in vibrotactile masking. Future research should examine the influence of frequency, intensity, and hand positioning to further improve the understanding of vibrotactile perception characteristics.

6 Conclusion

This study explored frequency-dependent masking in vibrotactile perception, analyzing how a masker influences the detectability of a maskee across different conditions. The masking thresholds were measured and evaluated using a psychophysical experiment with a three-alternative forced-choice (3AFC) method and an adaptive staircase process.

The results confirmed that masking effects vary by frequency but do not follow a strictly predictable pattern. Previous research states that maskers hide the nearby frequencies better, but individual differences played a meaningful role in this study. Some participants showed stronger masking effects at close frequencies, while others demonstrated more variability, highlighting the influence of mechanoreceptor sensitivity, adaptation effects, and cognitive factors. Additionally, threshold measurements were different among participants, underlining the importance of considering individual features in vibrotactile perception research. The repeated test for 200 Hz by the first participant showed that to check only three parameters is not enough for reliable conclusions. More frequency parameters and combinations could improve accuracy.

These findings will improve haptic feedback systems by optimizing frequency choice that can increase tactile signal transmission. Future research should include more participants, additional masker frequencies, and spatial variations like other fingers or places on the hand to refine the understanding of vibrotactile masking. Overall, this method provides valuable knowledge into vibrotactile perception and can contribute to improving haptic feedback technologies.

Bibliography

[1] Christophe Anet. “Auditory Masking and its Effect on our Perception of Sound”. In: (May 2021). URL: https://blogs.qsc.com/live-sound/auditory-masking-and-its-effect-on-our-perception-of-sound/.

[2] *Auditory masking*. 2024. URL: https://en.wikipedia.org/wiki/Auditory_masking.

[3] *Band-Limited White Noise*. 2024. URL: https://de.mathworks.com/help/simulink/slref/bandlimitedwhitenoise.html.

[4] *Band-pass filter*. URL: https://en.wikipedia.org/wiki/Band-pass_filter.

[5] Rahul Chaudhariu, Clemens Schuwerk, Mojtaba Danaei, Eckehard Steinbach. “Perceptual and Bitrate-Scalable Coding of Haptic Surface Texture Signals”. In: *IEEE Journal Of Selected Topics In Signal Processing* 9.3 (Apr. 2015), pp. 462–473.

[6] Richard D. Gilson. “Vibrotactile masking: Some spatial and temporal aspects”. In: *Perception & Psychophysics* 5 (May 1969), pp. 176–180. DOI: 10.3758/BF03209553.

[7] Marjorie R. Leek. “Adaptive procedures in psychophysical research”. In: *Perception & Psychophysics* 63 (Nov. 2001), pp. 1279–1292. DOI: 10.3758/BF03194543.

[8] Sebastian Merchel, M. Ercan Altinsoy. “Psychophysical comparison of the auditory and tactile perception: a survey”. In: *J Multimodal User Interfaces* 14 (July 2020), pp. 271–283. DOI: 10.1007/s12193-020-00333-z.

[9] A. Noll, L. Nockenberg, B. Gülecyüz, E. Steinbach. “VC-PWQ: Vibrotactile Signal Compression based on Perceptual Wavelet Quantization”. In: *IEEE World Haptics Conference (WHC)* (2021), pp. 427–432. DOI: 10.1109/WHC49131.2021.9517217.

[10] Seyed Ali Amirshahi Olga Cherepkova, Marius Pedersen. “A Comparative Analysis of the Three-Alternative Forced Choice Method and the Slider-Based Method in Subjective Experiments: A Case Study on Contrast Preference Task”. In: *SCITEPRESS – Science and Technology Publications, Lda* 3 (2024), pp. 425–435. DOI: 10.5220/0012360500003660.

[11] Andrew J Oxenham. “Mechanisms and mechanics of auditory masking”. In: *The Journal of Physiology* (May 2013). DOI: 10.1113/jphysiol.2013.254490.

[12] Miller Puckette. *Units of Amplitude*. 2006. URL: https://msp.ucsd.edu/techniques/v0.11/book-html/node9.html.

[13] *Root mean square*. URL: https://en.wikipedia.org/wiki/Root_mean_square.

[14] Patrick Timmons. “The basic of accelerometer calibration”. In: *PCB Piezotronics* (Apr. 2019).

[15] Ronald T. Verrillo, George A. Gescheider, Bruce G. Calman, Clayton L. van Doren. “Vibrotactile masking: Effects of one- and two-site stimulation”. In: (Jan. 1983), pp. 379–387.

[16] *Window function*. URL: https://en.wikipedia.org/wiki/Window_function.

[17] *Working with Decibels*. URL: https://learnemc.com/working-with-decibels.

Declaration

I hereby declare that the work presented in this thesis is entirely my own and that I did not use any other sources and references than the listed ones. I have marked all direct or indirect statements from other sources contained therein as quotations. Neither this work nor significant parts of it were part of another examination procedure. I have not published this work in whole or in part before. The electronic copy is consistent with all submitted copies.



Munich, 17.03.2025

place, date, signature